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Chain-of-Thought Prompting Elicits Reasoning in Large Language Models

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Abridged from the paper's version 6 (CC BY 4.0). Sections 3.3, 3.4 and 7 (Related Work) and the appendices are omitted, the figures are redrawn, and figure and table numbers follow the original.

ABSTRACT

We explore how generating a *chain of thought*—a series of intermediate reasoning steps—significantly improves the ability of large language models to perform complex reasoning. In particular, we show how such reasoning abilities emerge naturally in sufficiently large language models via a simple method called *chain-of-thought prompting*, where a few chain of thought demonstrations are provided as exemplars in prompting.

Experiments on three large language models show that chain-of-thought prompting improves performance on a range of arithmetic, commonsense, and symbolic reasoning tasks. The empirical gains can be striking. For instance, prompting a PaLM 540B with just eight chain-of-thought exemplars achieves state-of-the-art accuracy on the GSM8K benchmark of math word problems, surpassing even finetuned GPT-3 with a verifier.

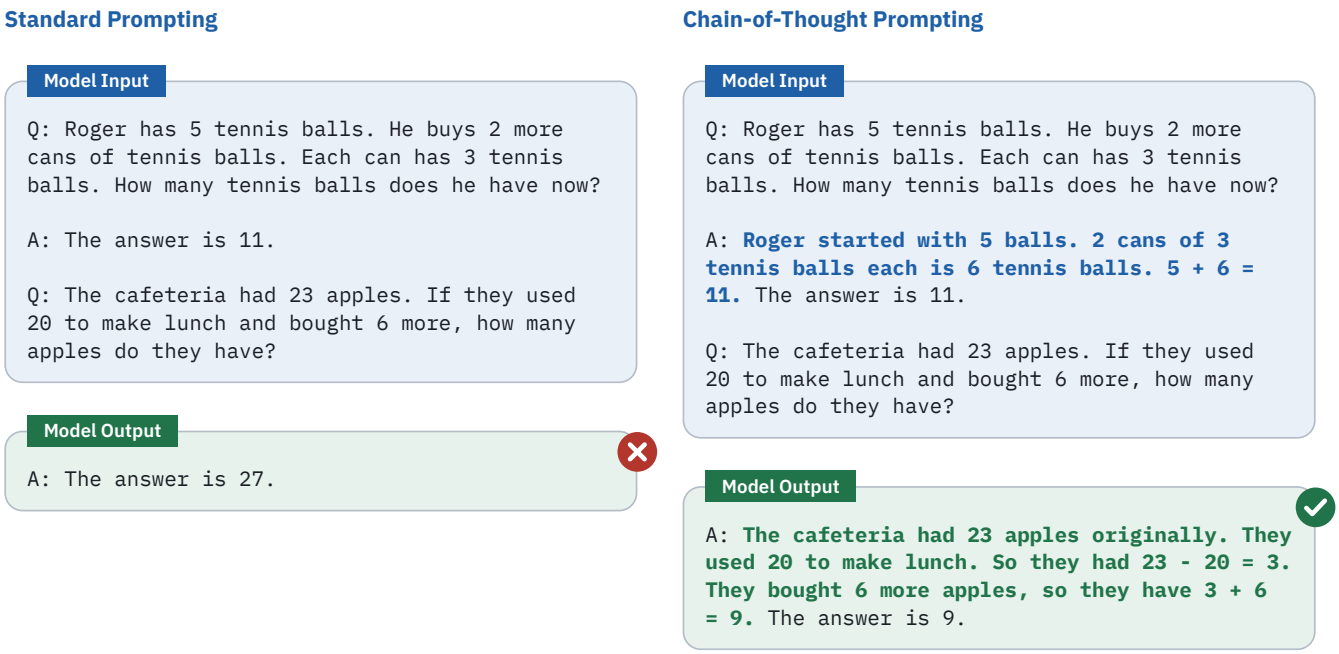


Figure 1. Chain-of-thought prompting enables large language models to tackle complex arithmetic, commonsense, and symbolic reasoning tasks. Chain-of-thought reasoning processes are highlighted.

1 Introduction

The NLP landscape has recently been revolutionized by language models (Peters et al., 2018; Devlin et al., 2019; Brown et al., 2020, *inter alia*). Scaling up the size of language models has been shown to confer a range of benefits, such as improved performance and sample efficiency (Kaplan et al., 2020; Brown et al., 2020, *inter alia*). However, scaling up model size alone has not proved sufficient for achieving high performance on challenging tasks such as arithmetic, commonsense, and symbolic reasoning (Rae et al., 2021).

This work explores how the reasoning ability of large language models can be unlocked by a simple method motivated by two ideas. First, techniques for arithmetic reasoning can benefit from generating natural language rationales that lead to the final answer. Prior work has given models the ability to generate natural language intermediate steps by training from scratch (Ling et al., 2017) or finetuning a pretrained model (Cobbe et al., 2021), in addition to neuro-symbolic methods that use formal languages instead of natural language (Roy and Roth, 2015; Chiang and Chen, 2019; Amini et al., 2019; X. Chen et al., 2019). Second, large language models offer the exciting prospect of in-context few-shot learning via *prompting*. That is, instead of finetuning a separate language model checkpoint for each new task, one can simply "prompt" the model with a few input–output exemplars demonstrating the task. Remarkably, this has been successful for a range of simple question-answering tasks (Brown et al., 2020).

Both of the above ideas, however, have key limitations. For rationale-augmented training and finetuning methods, it is costly to create a large set of high quality rationales, which is much more complicated than simple input–output pairs used in normal machine learning. For the traditional few-shot prompting method used in Brown et al. (2020), it works poorly on tasks that require reasoning abilities, and often does not improve substantially with increasing language model scale (Rae et al., 2021). In this paper, we combine the strengths of these two ideas in a way that avoids their limitations. Specifically, we explore the ability of language models to perform few-shot prompting for reasoning tasks, given a prompt that consists of triples: ‹input, *chain of thought*, output›. A *chain of thought* is a series of intermediate natural language reasoning steps that lead to the final output, and we refer to this approach as *chain-of-thought prompting*. An example prompt is shown in Figure 1.

We present empirical evaluations on arithmetic, commonsense, and symbolic reasoning benchmarks, showing that chain-of-thought prompting outperforms standard prompting, sometimes to a striking degree. Figure 2 illustrates one such result—on the GSM8K benchmark of math word problems (Cobbe et al., 2021), chain-of-thought prompting with PaLM 540B outperforms standard prompting by a large margin and achieves new state-of-the-art performance. A prompting only approach is important because it does not require a large training dataset and because a single model checkpoint can perform many tasks without loss of generality. This work underscores how large language models can learn via a few examples with natural language data about the task (c.f. automatically learning the patterns underlying inputs and outputs via a large training dataset).

2 Chain-of-Thought Prompting

Consider one's own thought process when solving a complicated reasoning task such as a multi-step math word problem. It is typical to decompose the problem into intermediate steps and solve each before giving the final answer: *"After Jane gives 2 flowers to her mom she has 10 ... then after she gives 3 to her dad she will have 7 ... so the answer is 7."* The goal of this paper is to endow language models with the ability to generate a similar *chain of thought*—a coherent series of intermediate reasoning steps that lead to the final answer for a problem. We will show that sufficiently large language models can generate chains of thought if demonstrations of chain-of-thought reasoning are provided in the exemplars for few-shot prompting.

Figure 1 shows an example of a model producing a chain of thought to solve a math word problem that it would have otherwise gotten incorrect. The chain of thought in this case resembles a solution and can interpreted as one, but we still opt to call it a chain of thought to better capture the idea that it mimics a step-by-step thought process for arriving at the answer (and also, solutions/explanations typically come *after* the final answer (Narang et al., 2020; Wiegreffe et al., 2022; Lampinen et al., 2022, *inter alia*)).

Chain-of-thought prompting has several attractive properties as an approach for facilitating reasoning in language models.

1. First, chain of thought, in principle, allows models to decompose multi-step problems into intermediate steps, which means that additional computation can be allocated to problems that require more reasoning steps.
2. Second, a chain of thought provides an interpretable window into the behavior of the model, suggesting how it might have arrived at a particular answer and providing opportunities to debug where the reasoning path went wrong (although fully characterizing a model's computations that support an answer remains an open question).
3. Third, chain-of-thought reasoning can be used for tasks such as math word problems, commonsense reasoning, and symbolic manipulation, and is potentially applicable (at least in principle) to any task that humans can solve via language.

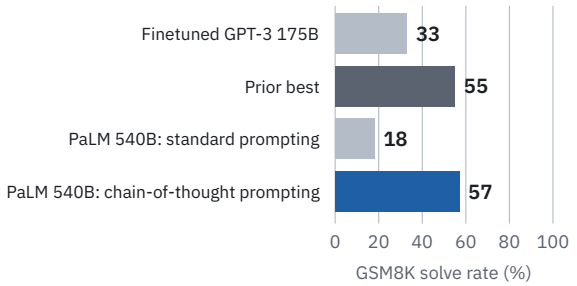


Figure 2. PaLM 540B uses chain-of-thought prompting to achieve new state-of-the-art performance on the GSM8K benchmark of math word problems. Finetuned GPT-3 and prior best are from Cobbe et al. (2021).

Redrawn from the values in the paper's Figure 2.

4. Finally, chain-of-thought reasoning can be readily elicited in sufficiently large off-the-shelf language models simply by including examples of chain of thought sequences into the exemplars of few-shot prompting.

In empirical experiments, we will observe the utility of chain-of-thought prompting for arithmetic reasoning (Section 3), commonsense reasoning (Section 4), and symbolic reasoning (Section 5).

3 Arithmetic Reasoning

We begin by considering math word problems of the form in Figure 1, which measure the arithmetic reasoning ability of language models. Though simple for humans, arithmetic reasoning is a task where language models often struggle (Hendrycks et al., 2021; Patel et al., 2021, *inter alia*). Strikingly, chain-of-thought prompting when used with the 540B parameter language model performs comparably with task-specific finetuned models on several tasks, even achieving new state of the art on the challenging GSM8K benchmark (Cobbe et al., 2021).

3.1 Experimental Setup

We explore chain-of-thought prompting for various language models on multiple benchmarks.

Benchmarks. We consider the following five math word problem benchmarks: **(1)** the **GSM8K** benchmark of math word problems (Cobbe et al., 2021), **(2)** the **SVAMP** dataset of math word problems with varying structures (Patel et al., 2021), **(3)** the **ASDiv** dataset of diverse math word problems (Miao et al., 2020), **(4)** the **AQuA** dataset of algebraic word problems, and **(5)** the **MAWPS** benchmark (Koncel-Kedziorski et al., 2016).

Standard prompting. For the baseline, we consider standard few-shot prompting, popularized by Brown et al. (2020), in which a language model is given in-context exemplars of input–output pairs before outputting a prediction for a test-time example. Exemplars are formatted as questions and answers. The model gives the answer directly, as shown in Figure 1 (left).

Chain-of-thought prompting. Our proposed approach is to augment each exemplar in few-shot prompting with a chain of thought for an associated answer, as illustrated in Figure 1 (right). As most of the datasets only have an evaluation split, we manually composed a set of eight few-shot exemplars with chains of thought for prompting—Figure 1 (right) shows one chain of thought exemplar. To investigate whether chain-of-thought prompting in this form can successfully elicit successful reasoning across a range of math word problems, we used this single set of eight chain of thought exemplars for all benchmarks except AQuA, which is multiple choice instead of free response.

Language models. We evaluate five large language models. The first is **GPT-3** (Brown et al., 2020), for which we use text-ada-001, text-babbage-001, text-curie-001, and text-davinci-002, which presumably correspond to InstructGPT models of 350M, 1.3B, 6.7B, and 175B parameters (Ouyang et al., 2022). The second is **LaMDA** (Thoppilan et al., 2022), which has models of 422M, 2B, 8B, 68B, and 137B parameters. The third is **PaLM**, which has models of 8B, 62B, and 540B parameters. The fourth is **UL2 20B** (Tay et al., 2022), and the fifth is **Codex** (M. Chen et al., 2021, code-davinci-002 in the OpenAI API). We sample from the models via greedy decoding (though follow-up work shows chain-of-thought prompting can be improved by taking the majority final answer over many sampled generations (Wang et al., 2022)). For LaMDA, we report averaged results over five random seeds, where each seed had a different randomly shuffled order of exemplars. As LaMDA experiments did not show large variance among different seeds, to save compute we report results for a single exemplar order for all other models.

3.2 Results

The strongest results of chain-of-thought prompting are summarized in Figure 4, with all experimental outputs for each model collection, model size, and benchmark shown in Table 2. There are three key takeaways. First, Figure 4 shows that chain-of-thought prompting is an emergent ability of model scale (Wei et al., 2022). That is, chain-of-thought prompting does not positively impact performance for small models, and only yields performance gains when used with models of ~100B parameters. We qualitatively found that models of smaller scale produced fluent but illogical chains of thought, leading to lower performance than standard prompting.

Second, chain-of-thought prompting has larger performance gains for more-complicated problems. For instance,

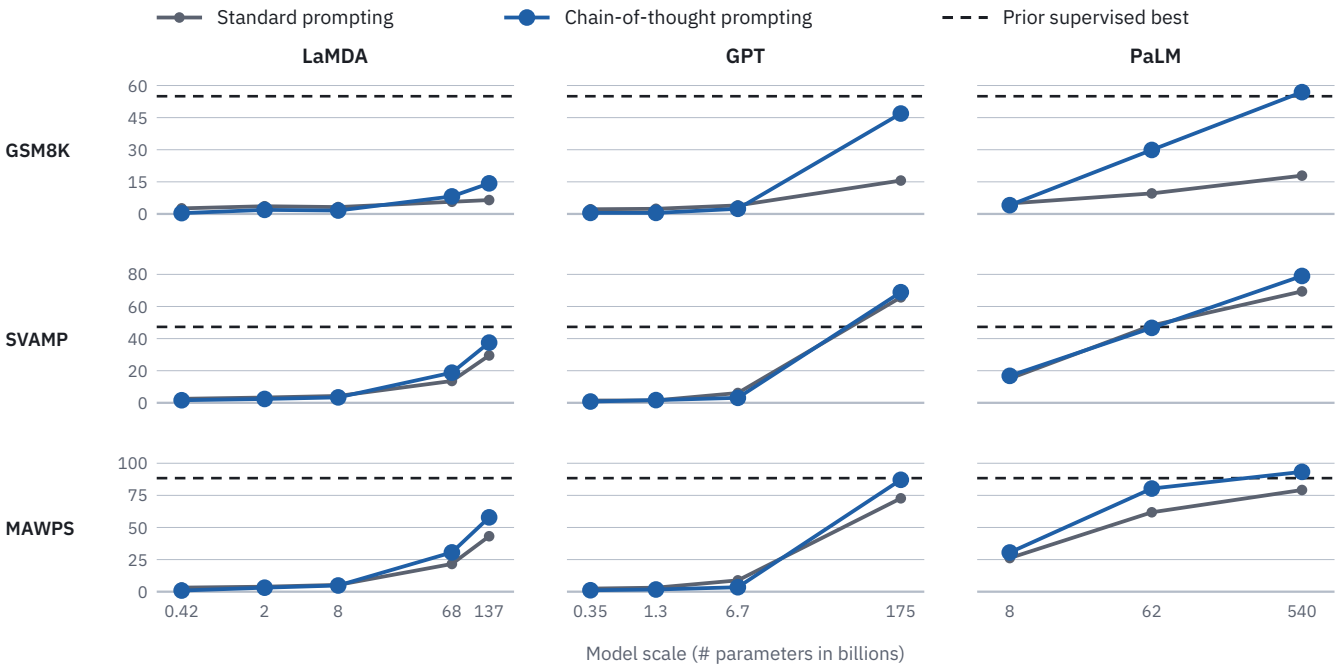


Figure 4. Chain-of-thought prompting enables large language models to solve challenging math problems. Notably, chain-of-thought reasoning is an emergent ability of increasing model scale. Prior best numbers are from Cobbe et al. (2021) for GSM8K, Jie et al. (2022) for SVAMP, and Lan et al. (2021) for MAWPS.

Redrawn from the solve rates in the paper's Table 2; the prior-best lines are those of its Figure 4.

for GSM8K (the dataset with the lowest baseline performance), performance more than doubled for the largest GPT and PaLM models.

Third, chain-of-thought prompting via GPT-3 175B and PaLM 540B compares favorably to prior state of the art, which typically finetunes a task-specific model on a labeled training dataset. Figure 4 shows how PaLM 540B uses chain-of-thought prompting to achieve new state of the art on GSM8K, SVAMP, and MAWPS (though note that standard prompting already passed the prior best for SVAMP). On the other two datasets, AQuA and ASDiv, PaLM with chain-of-thought prompting reaches within 2% of the state of the art.

To better understand why chain-of-thought prompting works, we manually examined model-generated chains of thought by LaMDA 137B for GSM8K. Of 50 random examples where the model returned the correct final answer, all of the generated chains of thought were also logically and mathematically correct except two that coincidentally arrived at the correct answer. We also randomly examined 50 random samples for which the model gave the wrong answer. The summary of this analysis is that 46% of the chains of thought were almost correct, barring minor mistakes (calculator error, symbol mapping error, or one reasoning step missing), and that the other 54% of the chains of thought had major errors in semantic understanding or coherence. To provide a small insight into why scaling improves chain-of-thought reasoning ability, we performed a similar analysis of errors made by PaLM 62B and whether those errors were fixed by scaling to PaLM 540B. The summary is that scaling PaLM to 540B fixes a large portion of one-step missing and semantic understanding errors in the 62B model.

4 Commonsense Reasoning

Although chain of thought is particularly suitable for math word problems, the language-based nature of chain of thought actually makes it applicable to a broad class of commonsense reasoning problems, which involve reasoning about physical and human interactions under the presumption of general background knowledge. Commonsense reasoning is key for interacting with the world and is still beyond the reach of current natural language understanding systems (Talmor et al., 2021).

Benchmarks. We consider five datasets covering a diverse range of commonsense reasoning types. The popular **CSQA** (Talmor et al., 2019) asks commonsense questions about the world involving complex semantics that often require prior knowledge. **StrategyQA** (Geva et al., 2021) requires models to infer a multi-hop strategy to answer questions. We choose two specialized evaluation sets from the BIG-bench effort (BIG-bench collaboration, 2021): **Date** Understanding, which involves inferring a date from a given context, and **Sports** Understanding, which involves determining whether a sentence relating to sports is plausible or implausible. Finally,

Table 2. Standard prompting versus chain of thought prompting on five arithmetic reasoning benchmarks. Note that chain of thought prompting is an emergent ability of model scale—it does not positively impact performance until used with a model of sufficient scale.

Model		GSM8K		SVAMP		ASDiv		AQuA		MAWPS	
		standard	CoT	standard	CoT	standard	CoT	standard	CoT	standard	CoT
UL2	20B	4.1	**4.4**	10.1	**12.5**	16.0	**16.9**	20.5	**23.6**	16.6	**19.1**
LaMDA	420M	2.6	0.4	2.5	1.6	3.2	0.8	23.5	8.3	3.2	0.9
	2B	3.6	1.9	3.3	2.4	4.1	3.8	22.9	17.7	3.9	3.1
	8B	3.2	1.6	4.3	3.4	5.9	5.0	22.8	18.6	5.3	4.8
	68B	5.7	**8.2**	13.6	**18.8**	21.8	**23.1**	22.3	20.2	21.6	**30.6**
	137B	6.5	**14.3**	29.5	**37.5**	40.1	**46.6**	25.5	20.6	43.2	**57.9**
GPT	350M	2.2	0.5	1.4	0.8	2.1	0.8	18.1	8.7	2.4	1.1
	1.3B	2.4	0.5	1.5	1.7	2.6	1.4	12.6	4.3	3.1	1.7
	6.7B	4.0	2.4	6.1	3.1	8.6	3.6	15.4	13.4	8.8	3.5
	175B	15.6	**46.9**	65.7	**68.9**	70.3	**71.3**	24.8	**35.8**	72.7	**87.1**
Codex	–	19.7	**63.1**	69.9	**76.4**	74.0	**80.4**	29.5	**45.3**	78.7	**92.6**
PaLM	8B	4.9	4.1	15.1	**16.8**	23.7	**25.2**	19.3	**21.7**	26.2	**30.5**
	62B	9.6	**29.9**	48.2	46.7	58.7	**61.9**	25.6	22.4	61.8	**80.3**
	540B	17.9	**56.9**	69.4	**79.0**	72.1	**73.9**	25.2	**35.8**	79.2	**93.3**

Solve rates (%). Bold: chain of thought above standard prompting.

the **SayCan** dataset (Ahn et al., 2022) involves mapping a natural language instruction to a sequence of robot actions from a discrete set.

Prompts. We follow the same experimental setup as the prior section. For CSQA and StrategyQA, we randomly selected examples from the training set and manually composed chains of thought for them to use as few-shot exemplars. The two BIG-bench tasks do not have training sets, so we selected the first ten examples as exemplars in the evaluation set as few-shot exemplars and report numbers on the rest of the evaluation set. For SayCan, we use six examples from the training set used in Ahn et al. (2022) and also manually composed chains of thought.

Results. For all tasks, scaling up model size improved the performance of standard prompting; chain-of-thought prompting led to further gains, with improvements appearing to be largest for PaLM 540B. With chain-of-thought prompting, PaLM 540B achieved strong performance relative to baselines, outperforming the prior state of the art on StrategyQA (75.6% vs 69.4%) and outperforming an unaided sports enthusiast on sports understanding (95.4% vs 84%). These results demonstrate that chain-of-thought prompting can also improve performance on tasks requiring a range of commonsense reasoning abilities (though note that gain was minimal on CSQA).

5 Symbolic Reasoning

Our final experimental evaluation considers symbolic reasoning, which is simple for humans but potentially challenging for language models. We show that chain-of-thought prompting not only enables language models to perform symbolic reasoning tasks that are challenging in the standard prompting setting, but also facilitates length generalization to inference-time inputs longer than those seen in the few-shot exemplars.

Tasks. We use the following two toy tasks.

- **Last letter concatenation.** This task asks the model to concatenate the last letters of words in a name. It is a more challenging version of first letter concatenation, which language models can already perform without chain of thought.[1] We generate full names by randomly concatenating names from the top one-thousand first and last names from name census data (https://namecensus.com/).
- **Coin flip.** This task asks the model to answer whether a coin is still heads up after people either flip or don't flip the coin.

As the construction of these symbolic reasoning tasks is well-defined, for each task we consider an *in-domain* test

[1] We tested 10 common names using GPT-3 davinci and it got all but one correct.

set for which examples had the same number of steps as the training/few-shot exemplars, as well as an *out-of-domain* (OOD) test set, for which evaluation examples had more steps than those in the exemplars. For last letter concatenation, the model only sees exemplars of names with two words, and then performs last letter concatenation on names with 3 and 4 words.[2] We do the same for the number of potential flips in the coin flip task. Our experimental setup uses the same methods and models as in the prior two sections. We again manually compose chains of thought for the few-shot exemplars for each task.

Results. Note that these in-domain evaluations are "toy tasks" in the sense that perfect solution structures are already provided by the chains of thought in the few-shot exemplars; all the model has to do is repeat the same steps with the new symbols in the test-time example. And yet, small models still fail—the ability to perform abstract manipulations on unseen symbols for these three tasks only arises at the scale of 100B model parameters.

As for the OOD evaluations, standard prompting fails for both tasks. With chain-of-thought prompting, language models achieve upward scaling curves (though performance is lower than in the in-domain setting). Hence, chain-of-thought prompting facilitates length generalization beyond seen chains of thought for language models of sufficient scale.

6 Discussion

We have explored chain-of-thought prompting as a simple mechanism for eliciting multi-step reasoning behavior in large language models. We first saw that chain-of-thought prompting improves performance by a large margin on arithmetic reasoning, yielding improvements that are much stronger than ablations and robust to different annotators, exemplars, and language models (Section 3). Next, experiments on commonsense reasoning underscored how the linguistic nature of chain-of-thought reasoning makes it generally applicable (Section 4). Finally, we showed that for symbolic reasoning, chain-of-thought prompting facilitates OOD generalization to longer sequence lengths (Section 5). In all experiments, chain-of-thought reasoning is elicited simply by prompting an off-the-shelf language model. No language models were finetuned in the process of writing this paper.

The emergence of chain-of-thought reasoning as a result of model scale has been a prevailing theme (Wei et al., 2022). For many reasoning tasks where standard prompting has a flat scaling curve, chain-of-thought prompting leads to dramatically increasing scaling curves. Chain-of-thought prompting appears to expand the set of tasks that large language models can perform successfully—in other words, our work underscores that standard prompting only provides a lower bound on the capabilities of large language models. This observation likely raises more questions than it answers—for instance, how much more can we expect reasoning ability to improve with a further increase in model scale? What other prompting methods might expand the range of tasks that language models can solve?

As for limitations, we first qualify that although chain of thought emulates the thought processes of human reasoners, this does not answer whether the neural network is actually "reasoning," which we leave as an open question. Second, although the cost of manually augmenting exemplars with chains of thought is minimal in the few-shot setting, such annotation costs could be prohibitive for finetuning (though this could potentially be surmounted with synthetic data generation, or zero-shot generalization). Third, there is no guarantee of correct reasoning paths, which can lead to both correct and incorrect answers; improving factual generations of language models is an open direction for future work (Rashkin et al., 2021; Ye and Durrett, 2022; Wiegreffe et al., 2022, *inter alia*). Finally, the emergence of chain-of-thought reasoning only at large model scales makes it costly to serve in real-world applications; further research could explore how to induce reasoning in smaller models.

7 Conclusions

We have explored chain-of-thought prompting as a simple and broadly applicable method for enhancing reasoning in language models. Through experiments on arithmetic, symbolic, and commonsense reasoning, we find that chain-of-thought reasoning is an emergent property of model scale that allows sufficiently large language models to perform reasoning tasks that otherwise have flat scaling curves. Broadening the range of reasoning tasks that language models can perform will hopefully inspire further work on language-based approaches to reasoning.

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[2] For names of length longer than 2 words, we concatenate multiple first and last names together.

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